Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Creekside Dairy

Physical address of dairy:

Number and Street	City	County	Zip Code
23234 Lone Tree RD	Escalon	San Joaquin	95320

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1957

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X229-X050-XX02-XXXX X229-X050-XX06-XXXX

B. OPERATORS

Dairy, Creekside

Operator name: Dairy, Creekside

Telephone no.: (209) 838-1180 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
23234 Lone Tree RD	Escalon	CA	95320

Van Vliet, Mike

Operator name: Van Vliet, Mike

Telephone no.: (209) 838-1180 (Landline) (209) 607-8343 (Cellular)

Mailing Address Number and Street	City	State	Zip Code
23234 Lone Tree RD	Escalon	CA	95320

This operator is responsible for paying permit fees.

C. OWNERS

Bros.,, Van Vliet

Legal owner name: Bros.,, Van Vliet

Telephone no.: (209) 838-1180 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
23234 Lone Tree RD	Escalon	CA	95320

Van Vliet, Mike

Legal owner name: Van Vliet, Mike		Telephone no.:	(209) 838-1180	(209) 607-8343
			Landline	Cellular
23234 Lone Tree RD	Escalon		CA	95320
Mailing Address Number and Street	City		State	Zip Code

This owner is responsible for paying permit fees.

AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	0	300	160	200
Number under roof	1,290	0	0	0	0	0
Maximum number	1,290	0	0	300	160	200
Average number	1,290	0	0	300	160	200
Avg live weight (lbs)	1,400	0	0	700		

Predominant milk cow breed: Holstein

Average milk production: 77 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 38,245.39 *tons per reporting period*

Total nitrogen from manure: 493,362.50 *lbs per reporting period*

After ammonia losses (30% loss applied): 345,353.75 *lbs per reporting period*

Total phosphorus from manure: 81,910.04 *lbs per reporting period*

Total potassium from manure: 252,979.61 *lbs per reporting period*

Total salt from manure: 607,396.50 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 13,680,000 *gallons*

Total nitrogen generated: 110,185.64 *lbs*

Total phosphorus generated: 8,398.66 *lbs*

Total potassium generated: 147,377.04 *lbs*

Total salt generated: 716,681.95 *lbs*

	13,680,000	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	13,680,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Canal	Ground water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

G. NUTRIENT EXPORTS

Date	Material type	Quantity	Reporting basis	Moisture (%)	Density (lbs/cu ft)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/08/2023	Separator solids	600.00 *ton*	Dry-weight	57.2		20,000.00	2,900.00	9,760.00		7.05

No liquid nutrient exports entered.

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Dry manure	10,272.00	1,489.44	5,012.74	36,208.80
Process wastewater	0.00	0.00	0.00	0.00
Total exports for all materials	10,272.00	1,489.44	5,012.74	36,208.80

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
10 Acre	17	17	2	manure	X229-X050-XX05-XXXX X229-X050-XX06-XXXX
Heifer	20	20	2	process wastewater	X229-X050-XX06-XXXX
Hutch	37	37	2	both	X229-X050-XX04-XXXX
L and M	32	32	2	process wastewater	X229-X050-XX30-XXXX
Leo's	29	29	2	process wastewater	X229-X050-XX05-XXXX
Morris	40	40	2	manure	X229-X050-XX08-XXXX
Pipe	27	27	2	manure	X229-X050-XX01-XXXX X229-X050-XX04-XXXX
Red Barn	33	33	2	process wastewater	X229-X050-XX01-XXXX
Ricks	25	25	2	process wastewater	X229-X050-XX02-XXXX X229-X050-XX04-XXXX
Sexton	24	24	2	both	X205-X240-XX03-XXXX
Shop	33	33	2	process wastewater	X229-X050-XX06-XXXX
Totals for areas that were used for application	317	317	22		
Totals for areas that were not used for application					
Land application area totals	317	317	22		

B. CROPS AND HARVESTS

10 Acre

Field name: 10 Acre

10/21/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 17 Plant date: 10/21/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/11/2023	240.00 *ton*	As-is		72.0	6,780.00	1,270.00	10,800.00		9.44

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	14.12	191.44	35.86	304.94	746.32

10 Acre

05/29/2023: Corn, silage

Crop: Corn, silage Acres planted: 17 Plant date: 05/29/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/13/2023	555.00 *ton*	As-is		68.4	4,080.00	1,140.00	6,070.00		6.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	32.00	256.00	48.00	211.20	0.00
Total actual harvest content	32.65	266.40	74.44	396.34	1,361.77

Heifer

Field name: Heifer

10/23/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 20 Plant date: 10/23/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/13/2023	315.00 *ton*	As-is		62.4	6,870.00	1,500.00	12,000.00		9.58

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	15.75	216.41	47.25	378.00	1,134.66

05/25/2023: Corn, silage

Crop: Corn, silage Acres planted: 20 Plant date: 05/25/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/09/2023	630.00 *ton*	As-is		66.4	4,470.00	720.00	5,960.00		7.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	32.00	256.00	48.00	211.20	0.00
Total actual harvest content	31.50	281.61	45.36	375.48	1,481.76

Hutch

Field name: Hutch

10/22/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 37 Plant date: 10/22/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/12/2023	555.00 *ton*	As-is		62.8	5,410.00	1,490.00	11,800.00		9.89

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	15.00	162.30	44.70	354.00	1,103.72

05/26/2023: Corn, silage

Crop: Corn, silage Acres planted: 37 Plant date: 05/26/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/12/2023	1,150.00 *ton*	As-is		70.4	4,110.00	550.00	5,290.00		6.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	32.00	256.00	48.00	211.20	0.00
Total actual harvest content	31.08	255.49	34.19	328.84	1,159.20

L and M

Field name: L and M

10/23/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 32 Plant date: 10/23/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/14/2023	480.00 *ton*	As-is		65.0	5,460.00	1,190.00	16,200.00		6.98

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	15.00	163.80	35.70	486.00	732.90

L and M

05/26/2023: Corn, silage

Crop: Corn, silage Acres planted: 32 Plant date: 05/26/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/14/2023	1,110.00 *ton*	As-is		66.6	4,540.00	710.00	6,150.00		6.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	32.00	256.00	48.00	211.20	0.00
Total actual harvest content	34.69	314.96	49.26	426.66	1,506.13

Leo's

Field name: Leo's

10/21/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 29 Plant date: 10/21/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/11/2023	470.00 *ton*	As-is		65.0	5,460.00	1,190.00	16,200.00		6.98

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	16.21	176.98	38.57	525.10	791.87

05/31/2023: Corn, silage

Crop: Corn, silage Acres planted: 29 Plant date: 05/31/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/14/2023	1,030.00 *ton*	As-is		68.6	4,050.00	1,020.00	5,960.00		6.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	32.00	256.00	48.00	211.20	0.00
Total actual harvest content	35.52	287.69	72.46	423.37	1,494.42

Morris

Field name: Morris

10/24/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 40 Plant date: 10/24/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/14/2023	520.00 *ton*	As-is		61.2	11,700.00	1,440.00	19,400.00		8.32

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	13.00	304.20	37.44	504.40	839.32

05/30/2023: Corn, silage

Crop: Corn, silage Acres planted: 40 Plant date: 05/30/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/12/2023	1,210.00 *ton*	As-is		66.8	4,180.00	760.00	4,660.00		6.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	32.00	256.00	48.00	211.20	0.00
Total actual harvest content	30.25	252.89	45.98	281.93	1,265.42

Pipe

Field name: Pipe

10/22/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 27 Plant date: 10/22/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/12/2023	345.00 *ton*	As-is		67.4	8,610.00	970.00	7,910.00		9.53

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	12.78	220.03	24.79	202.14	793.95

Pipe

05/27/2023: Corn, silage

Crop: Corn, silage Acres planted: 27 Plant date: 05/27/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/12/2023	850.00 *ton*	As-is		69.8	3,440.00	1,130.00	5,380.00		8.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	32.00	256.00	48.00	211.20	0.00
Total actual harvest content	31.48	216.59	71.15	338.74	1,521.19

Red Barn

Field name: Red Barn

10/22/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 33 Plant date: 10/22/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/10/2023	450.00 *ton*	As-is		64.2	6,730.00	1,040.00	14,200.00		12.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	13.64	183.55	28.36	387.27	1,191.16

05/29/2023: Corn, silage

Crop: Corn, silage Acres planted: 33 Plant date: 05/29/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/11/2023	1,100.00 *ton*	As-is		61.8	5,200.00	910.00	5,350.00		5.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	32.00	256.00	48.00	211.20	0.00
Total actual harvest content	33.33	346.67	60.67	356.67	1,375.20

Ricks

Field name: Ricks

10/20/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 25 Plant date: 10/20/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/11/2023	330.00 *ton*	As-is		67.4	8,610.00	970.00	7,910.00		9.53

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	13.20	227.30	25.61	208.82	820.19

05/30/2023: Corn, silage

Crop: Corn, silage Acres planted: 25 Plant date: 05/30/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/13/2023	700.00 *ton*	As-is		68.8	4,240.00	1,190.00	6,410.00		6.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	32.00	256.00	48.00	211.20	0.00
Total actual harvest content	28.00	237.44	66.64	358.96	1,135.68

Sexton

Field name: Sexton

10/20/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 24 Plant date: 10/20/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/10/2023	360.00 *ton*	As-is		67.4	4,320.00	1,280.00	10,600.00		9.97

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	15.00	129.60	38.40	318.00	975.07

Sexton

05/27/2023: Corn, silage

Crop: Corn, silage Acres planted: 24 Plant date: 05/27/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/11/2023	830.00 *ton*	As-is		70.6	4,060.00	660.00	4,390.00		5.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	32.00	256.00	48.00	211.20	0.00
Total actual harvest content	34.58	280.82	45.65	303.64	1,118.43

Shop

Field name: Shop

10/23/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 33 Plant date: 10/23/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/13/2023	495.00 *ton*	As-is		65.8	8,310.00	1,000.00	8,390.00		9.18

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	15.00	249.30	30.00	251.70	941.87

05/25/2023: Corn, silage

Crop: Corn, silage Acres planted: 33 Plant date: 05/25/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/09/2023	1,120.00 *ton*	As-is		69.4	4,100.00	580.00	4,760.00		6.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	32.00	256.00	48.00	211.20	0.00
Total actual harvest content	33.94	278.30	39.37	323.10	1,246.25

NUTRIENT BUDGET

A. LAND APPLICATIONS

10 Acre - 10/21/2022: Oats, silage-soft dough

Field name: 10 Acre

Crop: Oats, silage-soft dough — Plant date: 10/21/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/30/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	0.00	2,100,240.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/27/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	238.92	82.49	210.90	1,361.91	210.00 *ton*
Application event totals		238.92	82.49	210.90	1,361.91	

10 Acre - 05/29/2023: Corn, silage

Field name: 10 Acre

Crop: Corn, silage — Plant date: 05/29/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/05/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Separator Solids	Separator solids	277.88	89.20	298.46	1,406.54	240.00 *ton*
Application event totals		277.88	89.20	298.46	1,406.54	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	69.80	2,451,540.00 *gal*
Application event totals		0.00	0.00	0.00	69.80	

10 Acre - 05/29/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/29/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	8.00	0.00	0.00	0.00	
Application event totals		8.00	0.00	0.00	0.00	

07/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	69.80	2,451,540.00 *gal*
Application event totals		0.00	0.00	0.00	69.80	

07/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	49.86	1,751,100.00 *gal*
Application event totals		0.00	0.00	0.00	49.86	

07/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	49.86	1,751,100.00 *gal*
Application event totals		0.00	0.00	0.00	49.86	

07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	49.86	1,751,100.00 *gal*
Application event totals		0.00	0.00	0.00	49.86	

08/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	39.88	1,400,880.00 *gal*
Application event totals		0.00	0.00	0.00	39.88	

10 Acre - 05/29/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	49.86	1,751,100.00 *gal*
Application event totals		0.00	0.00	0.00	49.86	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	39.88	1,400,880.00 *gal*
Application event totals		0.00	0.00	0.00	39.88	

Heifer - 10/23/2022: Oats, silage-soft dough

Field name: Heifer

Crop: Oats, silage-soft dough Plant date: 10/23/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	283.90	19.91	420.59	1,682.35	504,000.00 *gal*
Canal	Ground water	0.00	0.00	0.00	0.00	2,450,280.00 *gal*
Application event totals		283.90	19.91	420.59	1,682.35	

Heifer - 05/25/2023: Corn, silage

Field name: Heifer

Crop: Corn, silage Plant date: 05/25/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Heifer - 05/25/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	59.33	2,451,540.00 *gal*
Application event totals		0.00	0.00	0.00	59.33	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/25/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	8.00	0.00	0.00	0.00	
Application event totals		8.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	59.33	2,451,540.00 *gal*
Application event totals		0.00	0.00	0.00	59.33	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	42.38	1,751,100.00 *gal*
Application event totals		0.00	0.00	0.00	42.38	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	143.42	10.05	203.23	1,040.96	396,000.00 *gal*
Canal	Ground water	0.00	0.00	0.00	25.10	1,037,190.00 *gal*
Application event totals		143.42	10.05	203.23	1,066.06	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	67.80	2,801,760.00 *gal*
Application event totals		0.00	0.00	0.00	67.80	

Heifer - 05/25/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	143.42	10.05	203.23	1,040.96	396,000.00 *gal*
Canal	Ground water	0.00	0.00	0.00	25.10	1,037,190.00 *gal*
Application event totals		143.42	10.05	203.23	1,066.06	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	67.80	2,801,760.00 *gal*
Application event totals		0.00	0.00	0.00	67.80	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	33.90	1,400,880.00 *gal*
Application event totals		0.00	0.00	0.00	33.90	

Hutch - 10/22/2022: Oats, silage-soft dough

Field name: Hutch

Crop: Oats, silage-soft dough

Plant date: 10/22/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/01/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	0.00	4,725,540.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/28/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Separator Solids	Separator solids	178.79	34.45	104.79	746.61	520.00 *ton*
Application event totals		178.79	34.45	104.79	746.61	

Hutch - 05/26/2023: Corn, silage

Field name: Hutch

Crop: Corn, silage

Plant date: 05/26/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	64.14	4,903,080.00 *gal*
Application event totals		0.00	0.00	0.00	64.14	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/26/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	8.00	0.00	0.00	0.00	
Application event totals		8.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	54.98	4,202,640.00 *gal*
Application event totals		0.00	0.00	0.00	54.98	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	24.67	1,885,800.00 *gal*
Application event totals		0.00	0.00	0.00	24.67	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	140.95	9.87	199.74	1,023.05	720,000.00 *gal*
Application event totals		140.95	9.87	199.74	1,023.05	

Hutch - 05/26/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	54.98	4,202,640.00 *gal*
Application event totals		0.00	0.00	0.00	54.98	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	140.95	9.87	199.74	1,023.05	720,000.00 *gal*
Canal	Ground water	0.00	0.00	0.00	24.67	1,885,800.00 *gal*
Application event totals		140.95	9.87	199.74	1,047.72	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	50.39	3,852,420.00 *gal*
Application event totals		0.00	0.00	0.00	50.39	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	48.10	3,677,310.00 *gal*
Application event totals		0.00	0.00	0.00	48.10	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	48.10	3,677,310.00 *gal*
Application event totals		0.00	0.00	0.00	48.10	

L and M - 10/23/2022: Oats, silage-soft dough

Field name: L and M

Crop: Oats, silage-soft dough

Plant date: 10/23/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

L and M - 10/23/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/02/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	60.89	4,025,460.00 *gal*
Application event totals		0.00	0.00	0.00	60.89	

02/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	218.09	22.24	204.05	1,040.77	828,000.00 *gal*
Application event totals		218.09	22.24	204.05	1,040.77	

L and M - 05/26/2023: Corn, silage

Field name: L and M

Crop: Corn, silage

Plant date: 05/26/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	63.57	4,202,640.00 *gal*
Application event totals		0.00	0.00	0.00	63.57	

05/26/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	8.00	0.00	0.00	0.00	
Application event totals		8.00	0.00	0.00	0.00	

06/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	58.27	3,852,420.00 *gal*
Application event totals		0.00	0.00	0.00	58.27	

L and M - 05/26/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	52.97	3,502,200.00 *gal*
Application event totals		0.00	0.00	0.00	52.97	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	25.67	1,697,220.00 *gal*
Application event totals		0.00	0.00	0.00	25.67	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	146.68	10.27	207.85	1,064.61	648,000.00 *gal*
Application event totals		146.68	10.27	207.85	1,064.61	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	55.62	3,677,310.00 *gal*
Application event totals		0.00	0.00	0.00	55.62	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	52.97	3,502,200.00 *gal*
Application event totals		0.00	0.00	0.00	52.97	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	138.53	9.70	196.31	1,005.47	612,000.00 *gal*
Canal	Ground water	0.00	0.00	0.00	24.24	1,602,930.00 *gal*
Application event totals		138.53	9.70	196.31	1,029.71	

L and M - 05/26/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	50.32	3,327,090.00 *gal*
Application event totals		0.00	0.00	0.00	50.32	

Leo's - 10/21/2022: Oats, silage-soft dough

Field name: Leo's

Crop: Oats, silage-soft dough

Plant date: 10/21/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	219.72	22.41	205.58	1,048.57	756,000.00 *gal*
Application event totals		219.72	22.41	205.58	1,048.57	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	0.00	3,675,420.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

Leo's - 05/31/2023: Corn, silage

Field name: Leo's

Crop: Corn, silage

Plant date: 05/31/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Leo's - 05/31/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	58.45	3,502,200.00 *gal*
Application event totals		0.00	0.00	0.00	58.45	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/31/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	8.00	0.00	0.00	0.00	
Application event totals		8.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	61.37	3,677,310.00 *gal*
Application event totals		0.00	0.00	0.00	61.37	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	55.53	3,327,090.00 *gal*
Application event totals		0.00	0.00	0.00	55.53	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	143.87	10.08	203.87	1,044.22	576,000.00 *gal*
Canal	Ground water	0.00	0.00	0.00	25.18	1,508,640.00 *gal*
Application event totals		143.87	10.08	203.87	1,069.40	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	52.61	3,151,980.00 *gal*
Application event totals		0.00	0.00	0.00	52.61	

Leo's - 05/31/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	52.61	3,151,980.00 *gal*
Application event totals		0.00	0.00	0.00	52.61	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	143.87	10.08	203.87	1,044.22	576,000.00 *gal*
Canal	Ground water	0.00	0.00	0.00	25.18	1,508,640.00 *gal*
Application event totals		143.87	10.08	203.87	1,069.40	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	49.68	2,976,870.00 *gal*
Application event totals		0.00	0.00	0.00	49.68	

Morris - 10/24/2022: Oats, silage-soft dough

Field name: Morris

Crop: Oats, silage-soft dough

Plant date: 10/24/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/27/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	236.93	81.81	209.14	1,350.56	490.00 *ton*
Application event totals		236.93	81.81	209.14	1,350.56	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/03/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	61.42	5,075,580.00 *gal*
Application event totals		0.00	0.00	0.00	61.42	

Morris - 05/30/2023: Corn, silage

Field name: Morris

Crop: Corn, silage

Plant date: 05/30/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/05/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Separator Solids	Separator solids	275.56	88.45	295.97	1,394.82	560.00 *ton*
Application event totals		275.56	88.45	295.97	1,394.82	

05/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	59.33	4,903,080.00 *gal*
Application event totals		0.00	0.00	0.00	59.33	

05/30/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	8.00	0.00	0.00	0.00	
Application event totals		8.00	0.00	0.00	0.00	

06/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	52.97	4,377,750.00 *gal*
Application event totals		0.00	0.00	0.00	52.97	

07/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	55.09	4,552,860.00 *gal*
Application event totals		0.00	0.00	0.00	55.09	

Morris - 05/30/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	52.97	4,377,750.00 *gal*
Application event totals		0.00	0.00	0.00	52.97	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	55.09	4,552,860.00 *gal*
Application event totals		0.00	0.00	0.00	55.09	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	57.21	4,727,970.00 *gal*
Application event totals		0.00	0.00	0.00	57.21	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	50.85	4,202,640.00 *gal*
Application event totals		0.00	0.00	0.00	50.85	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	50.85	4,202,640.00 *gal*
Application event totals		0.00	0.00	0.00	50.85	

Pipe - 10/22/2022: Oats, silage-soft dough

Field name: Pipe

Crop: Oats, silage-soft dough

Plant date: 10/22/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Pipe - 10/22/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/27/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	236.39	81.62	208.67	1,347.50	330.00 *ton*
Application event totals		236.39	81.62	208.67	1,347.50	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	0.00	3,325,380.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

Pipe - 05/27/2023: Corn, silage

Field name: Pipe

Crop: Corn, silage

Plant date: 05/27/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/05/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Separator Solids	Separator solids	277.02	88.92	297.54	1,402.20	380.00 *ton*
Application event totals		277.02	88.92	297.54	1,402.20	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	59.64	3,327,090.00 *gal*
Application event totals		0.00	0.00	0.00	59.64	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/27/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	8.00	0.00	0.00	0.00	
Application event totals		8.00	0.00	0.00	0.00	

Pipe - 05/27/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	56.50	3,151,980.00 *gal*
Application event totals		0.00	0.00	0.00	56.50	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	50.23	2,801,760.00 *gal*
Application event totals		0.00	0.00	0.00	50.23	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	53.36	2,976,870.00 *gal*
Application event totals		0.00	0.00	0.00	53.36	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	50.23	2,801,760.00 *gal*
Application event totals		0.00	0.00	0.00	50.23	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	50.23	2,801,760.00 *gal*
Application event totals		0.00	0.00	0.00	50.23	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	50.23	2,801,760.00 *gal*
Application event totals		0.00	0.00	0.00	50.23	

Pipe - 05/27/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	50.23	2,801,760.00 *gal*
Application event totals		0.00	0.00	0.00	50.23	

Red Barn - 10/22/2022: Oats, silage-soft dough

Field name: Red Barn

Crop: Oats, silage-soft dough

Plant date: 10/22/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/29/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	0.00	4,025,460.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

02/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	239.06	24.38	223.68	1,140.87	936,000.00 *gal*
Application event totals		239.06	24.38	223.68	1,140.87	

Red Barn - 05/29/2023: Corn, silage

Field name: Red Barn

Crop: Corn, silage

Plant date: 05/29/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Red Barn - 05/29/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	56.50	3,852,420.00 *gal*
Application event totals		0.00	0.00	0.00	56.50	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/29/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	8.00	0.00	0.00	0.00	
Application event totals		8.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	53.93	3,677,310.00 *gal*
Application event totals		0.00	0.00	0.00	53.93	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	51.37	3,502,200.00 *gal*
Application event totals		0.00	0.00	0.00	51.37	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	51.37	3,502,200.00 *gal*
Application event totals		0.00	0.00	0.00	51.37	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	142.24	9.96	201.55	1,032.35	648,000.00 *gal*
Canal	Ground water	0.00	0.00	0.00	24.89	1,697,220.00 *gal*
Application event totals		142.24	9.96	201.55	1,057.25	

Red Barn - 05/29/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	142.24	9.96	201.55	1,032.35	648,000.00 *gal*
Canal	Ground water	0.00	0.00	0.00	24.89	1,697,220.00 *gal*
Application event totals		142.24	9.96	201.55	1,057.25	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	48.80	3,327,090.00 *gal*
Application event totals		0.00	0.00	0.00	48.80	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	48.80	3,327,090.00 *gal*
Application event totals		0.00	0.00	0.00	48.80	

Ricks - 10/20/2022: Oats, silage-soft dough

Field name: Ricks

Crop: Oats, silage-soft dough

Plant date: 10/20/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/29/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	292.01	20.48	432.60	1,730.42	648,000.00 *gal*
Canal	Ground water	0.00	0.00	0.00	0.00	3,150,360.00 *gal*
Application event totals		292.01	20.48	432.60	1,730.42	

Ricks - 05/30/2023: Corn, silage

Ricks - 05/30/2023: Corn, silage

Field name: Ricks

Crop: Corn, silage

Plant date: 05/30/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	61.02	3,151,980.00 *gal*
Application event totals		0.00	0.00	0.00	61.02	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/30/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	8.00	0.00	0.00	0.00	
Application event totals		8.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	54.24	2,801,760.00 *gal*
Application event totals		0.00	0.00	0.00	54.24	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	50.85	2,626,650.00 *gal*
Application event totals		0.00	0.00	0.00	50.85	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	146.03	10.23	206.93	1,059.88	504,000.00 *gal*
Canal	Ground water	0.00	0.00	0.00	25.56	1,320,060.00 *gal*
Application event totals		146.03	10.23	206.93	1,085.44	

Ricks - 05/30/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	54.24	2,801,760.00 *gal*
Application event totals		0.00	0.00	0.00	54.24	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	135.60	9.50	192.15	984.18	468,000.00 *gal*
Canal	Ground water	0.00	0.00	0.00	23.73	1,225,770.00 *gal*
Application event totals		135.60	9.50	192.15	1,007.91	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	57.63	2,976,870.00 *gal*
Application event totals		0.00	0.00	0.00	57.63	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	54.24	2,801,760.00 *gal*
Application event totals		0.00	0.00	0.00	54.24	

Sexton - 10/20/2022: Oats, silage-soft dough

Field name: Sexton

Crop: Oats, silage-soft dough

Plant date: 10/20/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/30/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	0.00	2,975,340.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

Sexton - 10/20/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/28/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Separator Solids	Separator solids	174.92	33.70	102.52	730.46	330.00 *ton*
Application event totals		174.92	33.70	102.52	730.46	

Sexton - 05/27/2023: Corn, silage

Field name: Sexton

Crop: Corn, silage

Plant date: 05/27/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	63.57	3,151,980.00 *gal*
Application event totals		0.00	0.00	0.00	63.57	

05/27/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	8.00	0.00	0.00	0.00	
Application event totals		8.00	0.00	0.00	0.00	

07/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	60.03	2,976,870.00 *gal*
Application event totals		0.00	0.00	0.00	60.03	

07/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	56.50	2,801,760.00 *gal*
Application event totals		0.00	0.00	0.00	56.50	

Sexton - 05/27/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	52.97	2,626,650.00 *gal*
Application event totals		0.00	0.00	0.00	52.97	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	152.11	10.65	215.55	1,104.04	504,000.00 *gal*
Canal	Ground water	0.00	0.00	0.00	26.62	1,320,060.00 *gal*
Application event totals		152.11	10.65	215.55	1,130.67	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	52.97	2,626,650.00 *gal*
Application event totals		0.00	0.00	0.00	52.97	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	130.38	9.13	184.76	946.32	432,000.00 *gal*
Canal	Ground water	0.00	0.00	0.00	22.82	1,131,480.00 *gal*
Application event totals		130.38	9.13	184.76	969.14	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	49.44	2,451,540.00 *gal*
Application event totals		0.00	0.00	0.00	49.44	

Shop - 10/23/2022: Oats, silage-soft dough

Field name: Shop

Crop: Oats, silage-soft dough

Plant date: 10/23/2022

Shop - 10/23/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/02/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	294.96	20.69	436.97	1,747.90	864,000.00 *gal*
Canal	Ground water	0.00	0.00	0.00	0.00	4,200,480.00 *gal*
Application event totals		294.96	20.69	436.97	1,747.90	

Shop - 05/25/2023: Corn, silage

Field name: Shop

Crop: Corn, silage

Plant date: 05/25/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	59.07	4,027,530.00 *gal*
Application event totals		0.00	0.00	0.00	59.07	

05/25/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	8.00	0.00	0.00	0.00	
Application event totals		8.00	0.00	0.00	0.00	

06/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	56.50	3,852,420.00 *gal*
Application event totals		0.00	0.00	0.00	56.50	

07/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	53.93	3,677,310.00 *gal*
Application event totals		0.00	0.00	0.00	53.93	

Shop - 05/25/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	53.93	3,677,310.00 *gal*
Application event totals		0.00	0.00	0.00	53.93	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	142.24	9.96	201.55	1,032.35	648,000.00 *gal*
Canal	Ground water	0.00	0.00	0.00	24.89	1,697,220.00 *gal*
Application event totals		142.24	9.96	201.55	1,057.25	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	142.24	9.96	201.55	1,032.35	648,000.00 *gal*
Canal	Ground water	0.00	0.00	0.00	24.89	1,697,220.00 *gal*
Application event totals		142.24	9.96	201.55	1,057.25	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	51.37	3,502,200.00 *gal*
Application event totals		0.00	0.00	0.00	51.37	

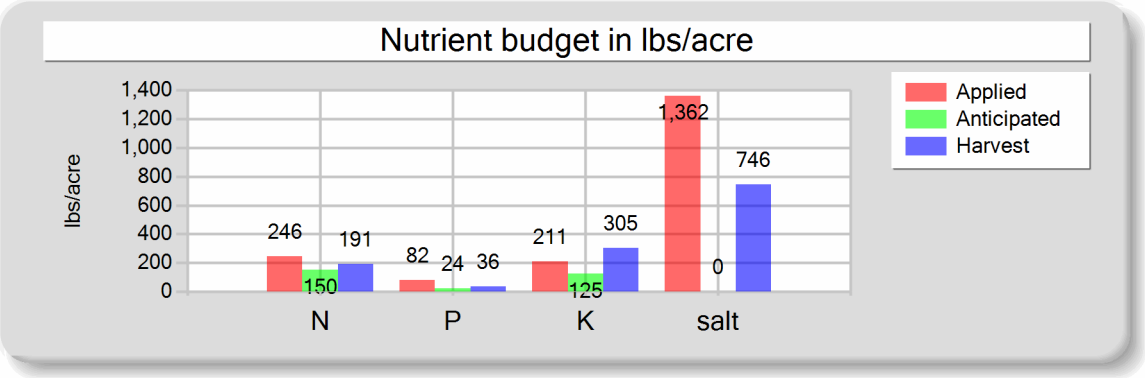
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	50.34	3,432,156.00 *gal*
Application event totals		0.00	0.00	0.00	50.34	

B. NUTRIENT BUDGET

10 Acre - 10/21/2022: Oats, silage-soft dough

Field name: 10 Acre | Crop: Oats, silage-soft dough | Plant date: 10/21/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	238.92	82.49	210.90	1,361.91
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	245.92	82.49	210.90	1,361.91
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	191.44	35.86	304.94	746.32
Nutrient balance	54.48	46.63	-94.04	615.60
Applied to removed ratio	1.28	2.30	0.69	1.82

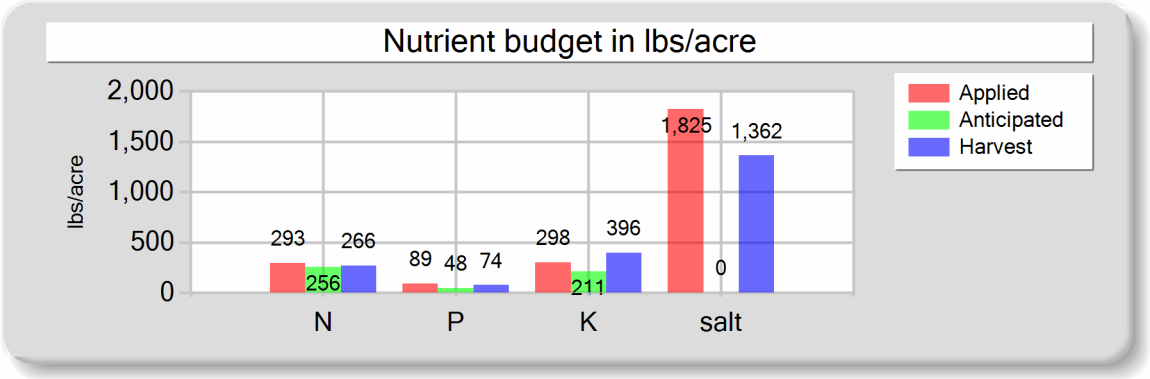
Fresh water applied	
2,100,240.00	*gallons*
77.34	*acre-inches*
4.55	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

10 Acre - 05/29/2023: Corn, silage

Field name: 10 Acre | Crop: Corn, silage | Plant date: 05/29/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	8.00	0.00	0.00	0.00
Dry manure	277.88	89.20	298.46	1,406.54
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	418.79
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	292.88	89.20	298.46	1,825.33
Anticipated crop nutrient removal	256.00	48.00	211.20	0.00
Actual crop nutrient removal	266.40	74.44	396.34	1,361.77
Nutrient balance	26.48	14.76	-97.87	463.56
Applied to removed ratio	1.10	1.20	0.75	1.34

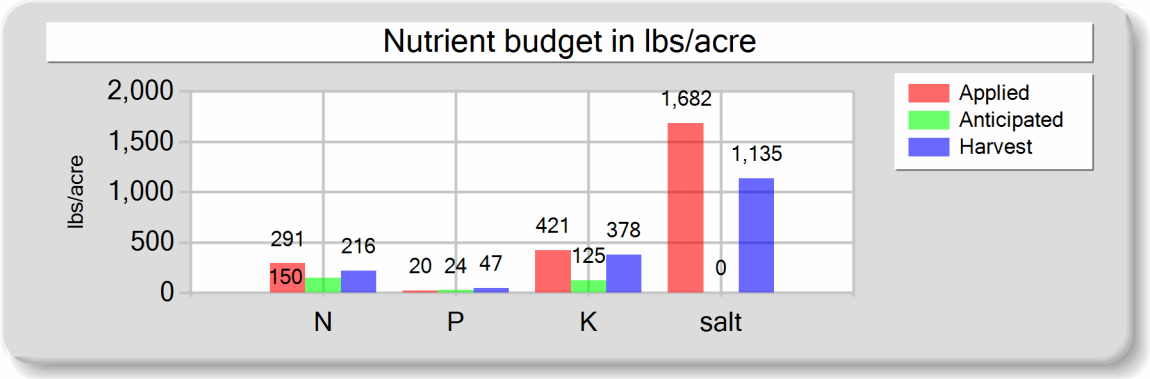
Fresh water applied	
14,709,240.00	*gallons*
541.69	*acre-inches*
31.86	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Heifer - 10/23/2022: Oats, silage-soft dough

Field name: Heifer | Crop: Oats, silage-soft dough | Plant date: 10/23/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	283.90	19.91	420.59	1,682.35
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	290.90	19.91	420.59	1,682.35
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	216.41	47.25	378.00	1,134.66
Nutrient balance	74.49	-27.34	42.59	547.70
Applied to removed ratio	1.34	0.42	1.11	1.48

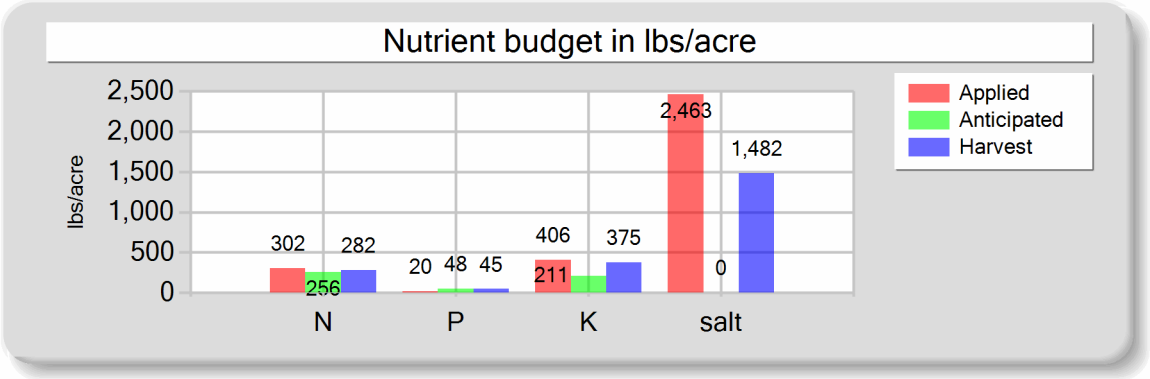
Fresh water applied	
2,450,280.00	*gallons*
90.24	*acre-inches*
4.51	*inches/acre*

Process wastewater applied	
504,000.00	*gallons*
18.56	*acre-inches*
0.93	*inches/acre*

Total harvests for the crop	
1	*harvests*

Heifer - 05/25/2023: Corn, silage

Field name: Heifer | Crop: Corn, silage | Plant date: 05/25/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	8.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	286.84	20.09	406.47	2,081.91
Fresh water	0.00	0.00	0.00	380.75
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	301.84	20.09	406.47	2,462.66
Anticipated crop nutrient removal	256.00	48.00	211.20	0.00
Actual crop nutrient removal	281.61	45.36	375.48	1,481.76
Nutrient balance	20.23	-25.27	30.99	980.90
Applied to removed ratio	1.07	0.44	1.08	1.66

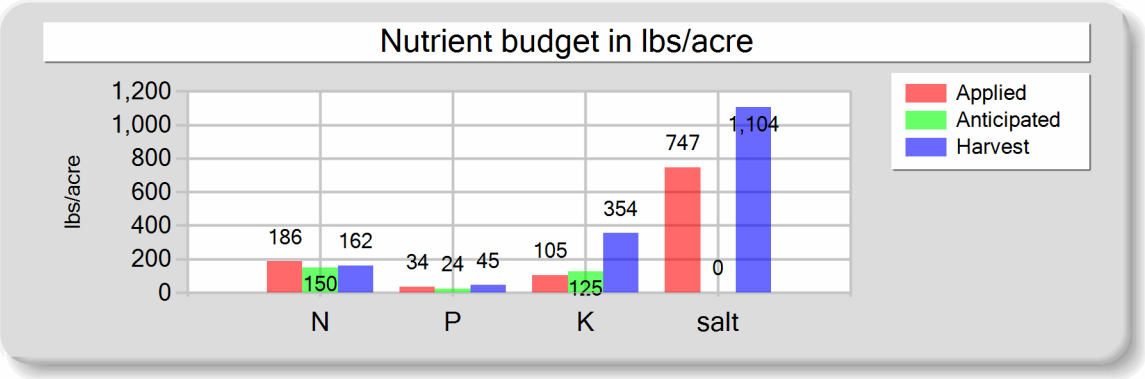
Fresh water applied	
15,732,960.00	*gallons*
579.39	*acre-inches*
28.97	*inches/acre*

Process wastewater applied	
792,000.00	*gallons*
29.17	*acre-inches*
1.46	*inches/acre*

Total harvests for the crop	
1	*harvests*

Hutch - 10/22/2022: Oats, silage-soft dough

Field name: Hutch | Crop: Oats, silage-soft dough | Plant date: 10/22/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	178.79	34.45	104.79	746.61
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	185.79	34.45	104.79	746.61
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	162.30	44.70	354.00	1,103.72
Nutrient balance	23.49	-10.25	-249.21	-357.12
Applied to removed ratio	1.14	0.77	0.30	0.68

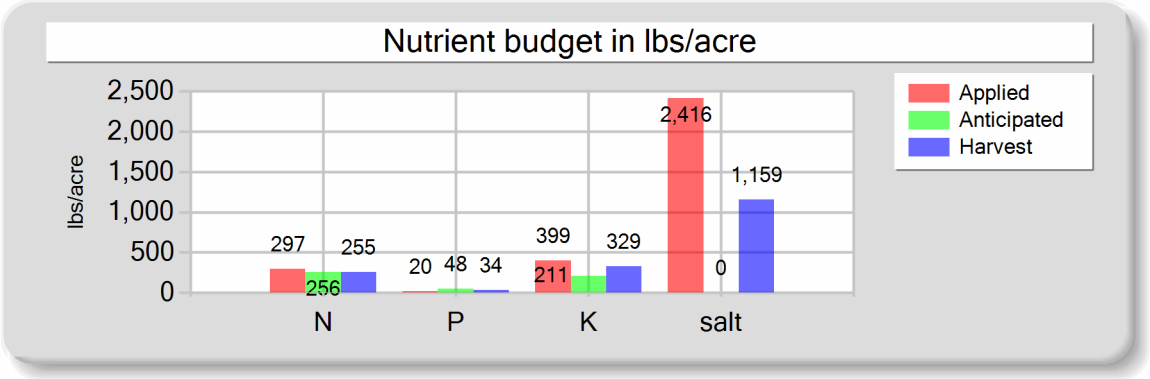
Fresh water applied	
4,725,540.00	*gallons*
174.03	*acre-inches*
4.70	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Hutch - 05/26/2023: Corn, silage

Field name: Hutch | Crop: Corn, silage | Plant date: 05/26/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	8.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	281.91	19.75	399.48	2,046.10
Fresh water	0.00	0.00	0.00	370.03
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	296.91	19.75	399.48	2,416.14
Anticipated crop nutrient removal	256.00	48.00	211.20	0.00
Actual crop nutrient removal	255.49	34.19	328.84	1,159.20
Nutrient balance	41.42	-14.44	70.64	1,256.94
Applied to removed ratio	1.16	0.58	1.21	2.08

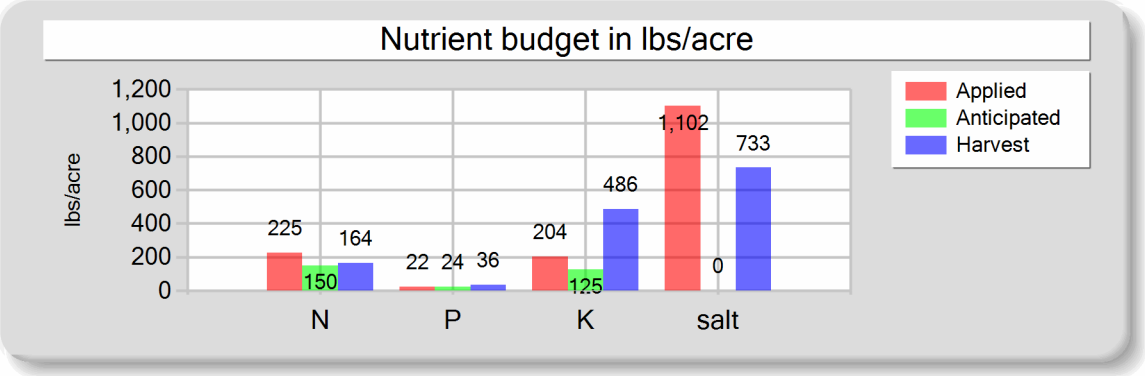
Fresh water applied	
28,287,000.00	*gallons*
1,041.71	*acre-inches*
28.15	*inches/acre*

Process wastewater applied	
1,440,000.00	*gallons*
53.03	*acre-inches*
1.43	*inches/acre*

Total harvests for the crop	
1	*harvests*

L and M - 10/23/2022: Oats, silage-soft dough

Field name: L and M Crop: Oats, silage-soft dough Plant date: 10/23/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	218.09	22.24	204.05	1,040.77
Fresh water	0.00	0.00	0.00	60.89
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	225.09	22.24	204.05	1,101.65
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	163.80	35.70	486.00	732.90
Nutrient balance	61.29	-13.46	-281.95	368.75
Applied to removed ratio	1.37	0.62	0.42	1.50

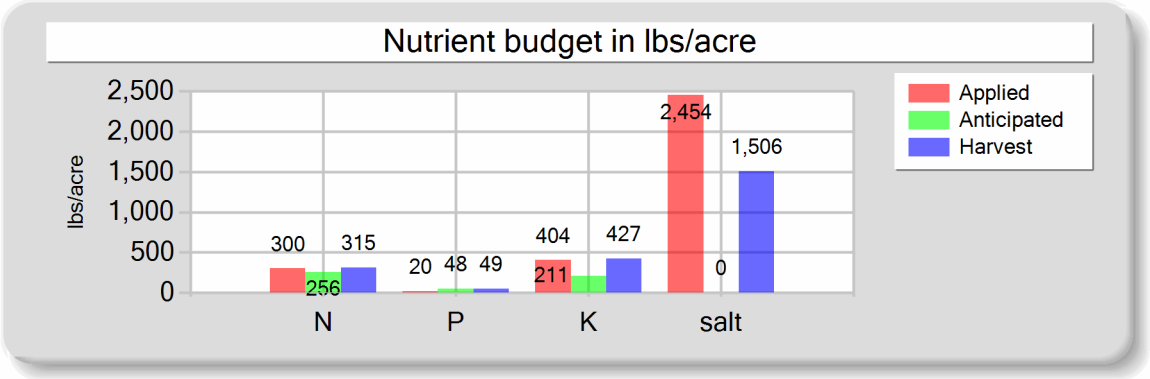
Fresh water applied	
4,025,460.00	*gallons*
148.24	*acre-inches*
4.63	*inches/acre*

Process wastewater applied	
828,000.00	*gallons*
30.49	*acre-inches*
0.95	*inches/acre*

Total harvests for the crop	
1	*harvests*

L and M - 05/26/2023: Corn, silage

Field name: L and M Crop: Corn, silage Plant date: 05/26/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	8.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	285.21	19.98	404.16	2,070.08
Fresh water	0.00	0.00	0.00	383.64
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	300.21	19.98	404.16	2,453.72
Anticipated crop nutrient removal	256.00	48.00	211.20	0.00
Actual crop nutrient removal	314.96	49.26	426.66	1,506.13
Nutrient balance	-14.75	-29.28	-22.50	947.59
Applied to removed ratio	0.95	0.41	0.95	1.63

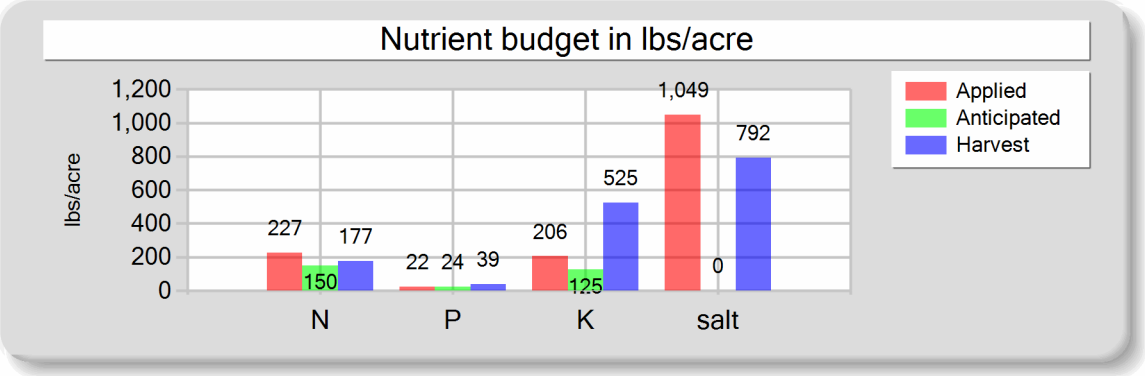
Fresh water applied	
25,364,010.00	*gallons*
934.07	*acre-inches*
29.19	*inches/acre*

Process wastewater applied	
1,260,000.00	*gallons*
46.40	*acre-inches*
1.45	*inches/acre*

Total harvests for the crop	
1	*harvests*

Leo's - 10/21/2022: Oats, silage-soft dough

Field name: Leo's | Crop: Oats, silage-soft dough | Plant date: 10/21/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	219.72	22.41	205.58	1,048.57
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	226.72	22.41	205.58	1,048.57
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	176.98	38.57	525.10	791.87
Nutrient balance	49.74	-16.17	-319.52	256.70
Applied to removed ratio	1.28	0.58	0.39	1.32

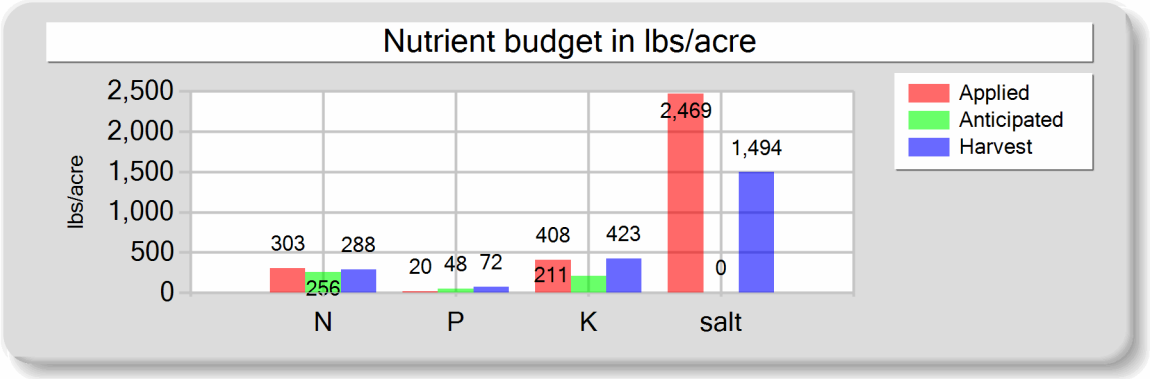
Fresh water applied	
3,675,420.00	*gallons*
135.35	*acre-inches*
4.67	*inches/acre*

Process wastewater applied	
756,000.00	*gallons*
27.84	*acre-inches*
0.96	*inches/acre*

Total harvests for the crop	
1	*harvests*

Leo's - 05/31/2023: Corn, silage

Field name: Leo's Crop: Corn, silage Plant date: 05/31/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	8.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	287.74	20.16	407.74	2,088.44
Fresh water	0.00	0.00	0.00	380.61
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	302.74	20.16	407.74	2,469.05
Anticipated crop nutrient removal	256.00	48.00	211.20	0.00
Actual crop nutrient removal	287.69	72.46	423.37	1,494.42
Nutrient balance	15.05	-52.30	-15.62	974.62
Applied to removed ratio	1.05	0.28	0.96	1.65

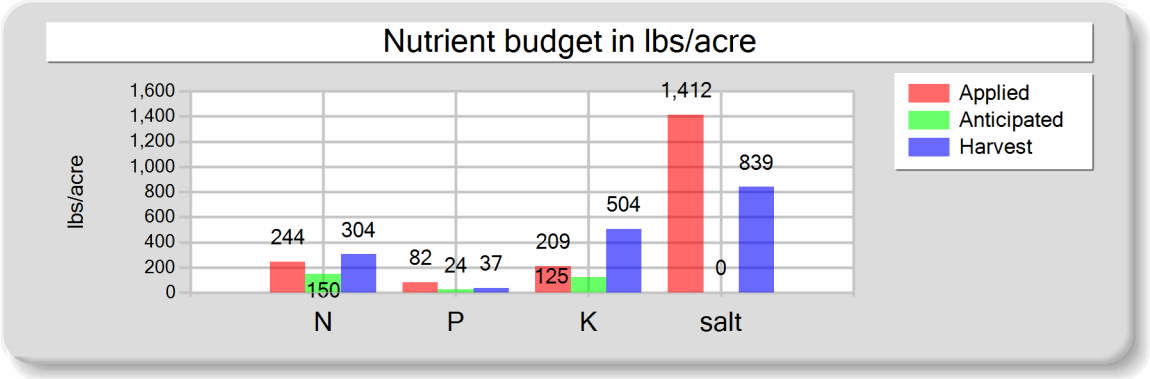
Fresh water applied	
22,804,710.00	*gallons*
839.82	*acre-inches*
28.96	*inches/acre*

Process wastewater applied	
1,152,000.00	*gallons*
42.42	*acre-inches*
1.46	*inches/acre*

Total harvests for the crop	
1	*harvests*

Morris - 10/24/2022: Oats, silage-soft dough

Field name: Morris | Crop: Oats, silage-soft dough | Plant date: 10/24/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	236.93	81.81	209.14	1,350.56
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	61.42
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	243.93	81.81	209.14	1,411.98
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	304.20	37.44	504.40	839.32
Nutrient balance	-60.27	44.37	-295.26	572.66
Applied to removed ratio	0.80	2.18	0.41	1.68

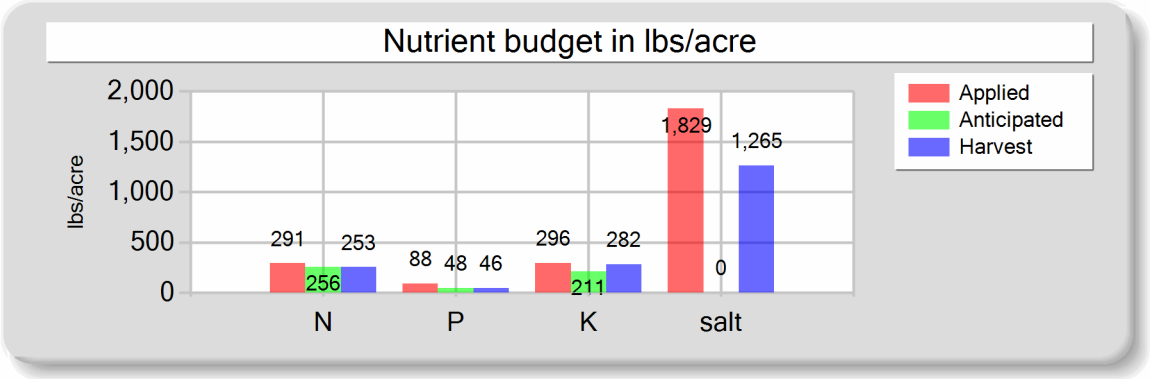
Fresh water applied	
5,075,580.00	*gallons*
186.92	*acre-inches*
4.67	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Morris - 05/30/2023: Corn, silage

Field name: Morris | Crop: Corn, silage | Plant date: 05/30/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	8.00	0.00	0.00	0.00
Dry manure	275.56	88.45	295.97	1,394.82
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	434.37
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	290.56	88.45	295.97	1,829.19
Anticipated crop nutrient removal	256.00	48.00	211.20	0.00
Actual crop nutrient removal	252.89	45.98	281.93	1,265.42
Nutrient balance	37.67	42.47	14.04	563.77
Applied to removed ratio	1.15	1.92	1.05	1.45

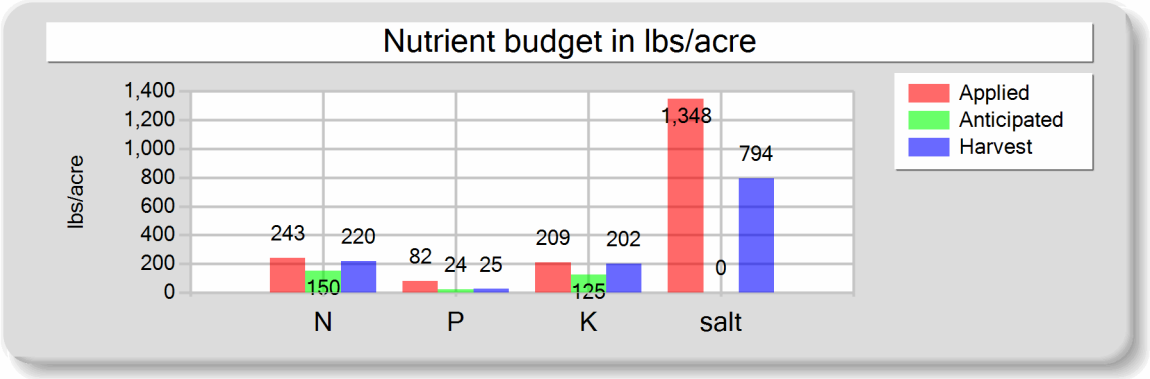
Fresh water applied	
35,897,550.00	*gallons*
1,321.98	*acre-inches*
33.05	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Pipe - 10/22/2022: Oats, silage-soft dough

Field name: Pipe | Crop: Oats, silage-soft dough | Plant date: 10/22/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	236.39	81.62	208.67	1,347.50
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	243.39	81.62	208.67	1,347.50
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	220.03	24.79	202.14	793.95
Nutrient balance	23.36	56.83	6.53	553.55
Applied to removed ratio	1.11	3.29	1.03	1.70

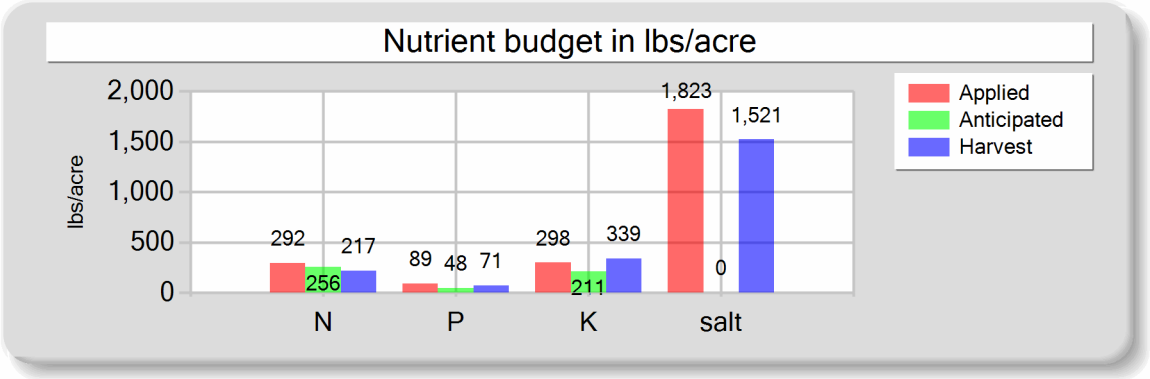
Fresh water applied	
3,325,380.00	*gallons*
122.46	*acre-inches*
4.54	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Pipe - 05/27/2023: Corn, silage

Field name: Pipe | Crop: Corn, silage | Plant date: 05/27/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	8.00	0.00	0.00	0.00
Dry manure	277.02	88.92	297.54	1,402.20
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	420.64
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	292.02	88.92	297.54	1,822.84
Anticipated crop nutrient removal	256.00	48.00	211.20	0.00
Actual crop nutrient removal	216.59	71.15	338.74	1,521.19
Nutrient balance	75.43	17.77	-41.20	301.65
Applied to removed ratio	1.35	1.25	0.88	1.20

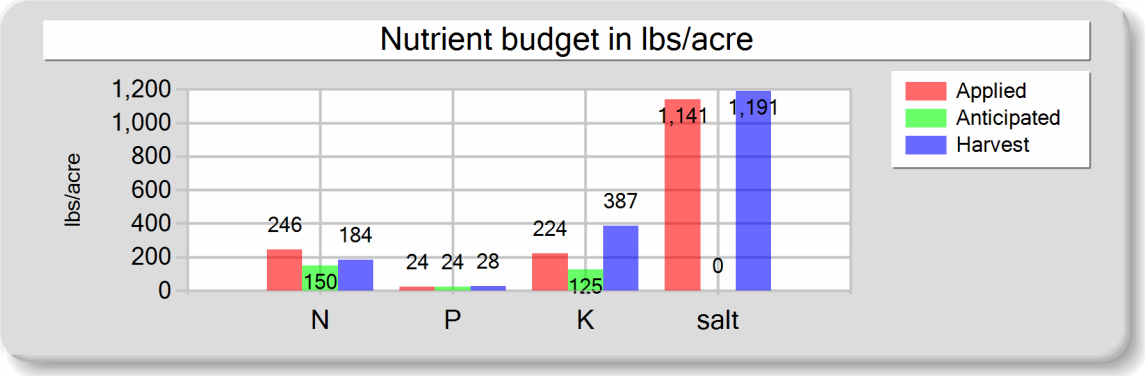
Fresh water applied	
23,464,740.00	*gallons*
864.13	*acre-inches*
32.00	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Red Barn - 10/22/2022: Oats, silage-soft dough

Field name: Red Barn | Crop: Oats, silage-soft dough | Plant date: 10/22/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	239.06	24.38	223.68	1,140.87
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	246.06	24.38	223.68	1,140.87
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	183.55	28.36	387.27	1,191.16
Nutrient balance	62.52	-3.98	-163.60	-50.30
Applied to removed ratio	1.34	0.86	0.58	0.96

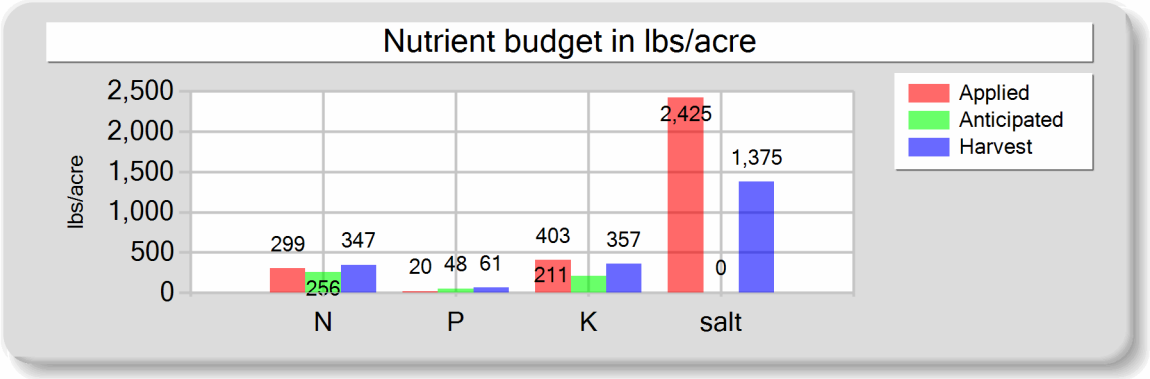
Fresh water applied	
4,025,460.00	*gallons*
148.24	*acre-inches*
4.49	*inches/acre*

Process wastewater applied	
936,000.00	*gallons*
34.47	*acre-inches*
1.04	*inches/acre*

Total harvests for the crop	
1	*harvests*

Red Barn - 05/29/2023: Corn, silage

Field name: Red Barn Crop: Corn, silage Plant date: 05/29/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	8.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	284.47	19.93	403.11	2,064.70
Fresh water	0.00	0.00	0.00	360.55
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	299.47	19.93	403.11	2,425.26
Anticipated crop nutrient removal	256.00	48.00	211.20	0.00
Actual crop nutrient removal	346.67	60.67	356.67	1,375.20
Nutrient balance	-47.20	-40.74	46.44	1,050.06
Applied to removed ratio	0.86	0.33	1.13	1.76

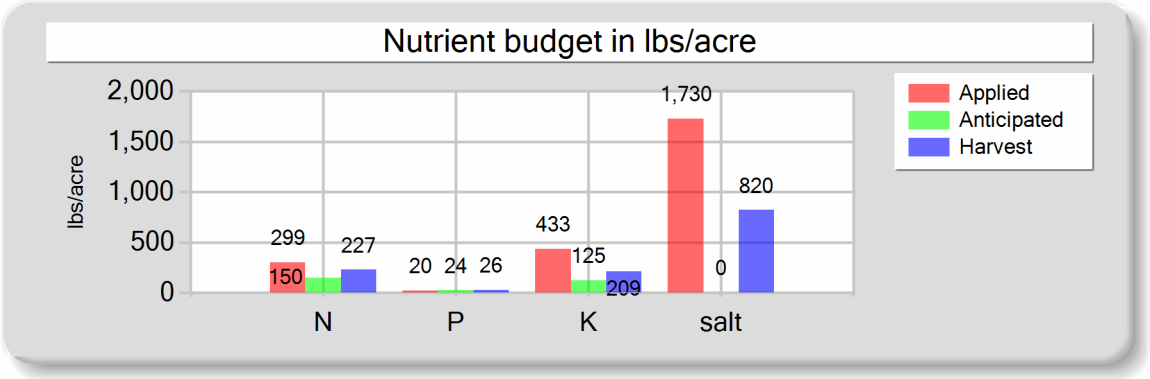
Fresh water applied	
24,582,750.00	*gallons*
905.30	*acre-inches*
27.43	*inches/acre*

Process wastewater applied	
1,296,000.00	*gallons*
47.73	*acre-inches*
1.45	*inches/acre*

Total harvests for the crop	
1	*harvests*

Ricks - 10/20/2022: Oats, silage-soft dough

Field name: Ricks | Crop: Oats, silage-soft dough | Plant date: 10/20/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	292.01	20.48	432.60	1,730.42
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	299.01	20.48	432.60	1,730.42
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	227.30	25.61	208.82	820.19
Nutrient balance	71.70	-5.12	223.78	910.23
Applied to removed ratio	1.32	0.80	2.07	2.11

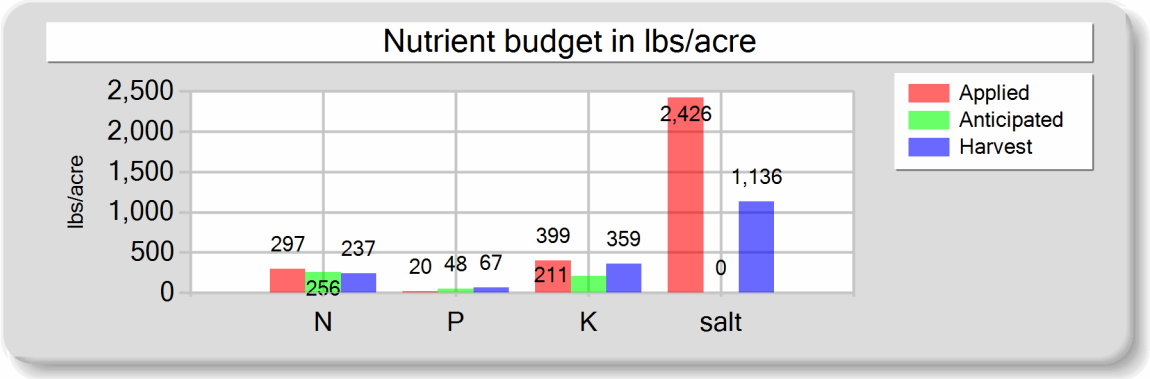
Fresh water applied	
3,150,360.00	*gallons*
116.02	*acre-inches*
4.64	*inches/acre*

Process wastewater applied	
648,000.00	*gallons*
23.86	*acre-inches*
0.95	*inches/acre*

Total harvests for the crop	
1	*harvests*

Ricks - 05/30/2023: Corn, silage

Field name: Ricks | Crop: Corn, silage | Plant date: 05/30/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	8.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	281.63	19.73	399.08	2,044.06
Fresh water	0.00	0.00	0.00	381.53
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	296.63	19.73	399.08	2,425.59
Anticipated crop nutrient removal	256.00	48.00	211.20	0.00
Actual crop nutrient removal	237.44	66.64	358.96	1,135.68
Nutrient balance	59.19	-46.91	40.12	1,289.91
Applied to removed ratio	1.25	0.30	1.11	2.14

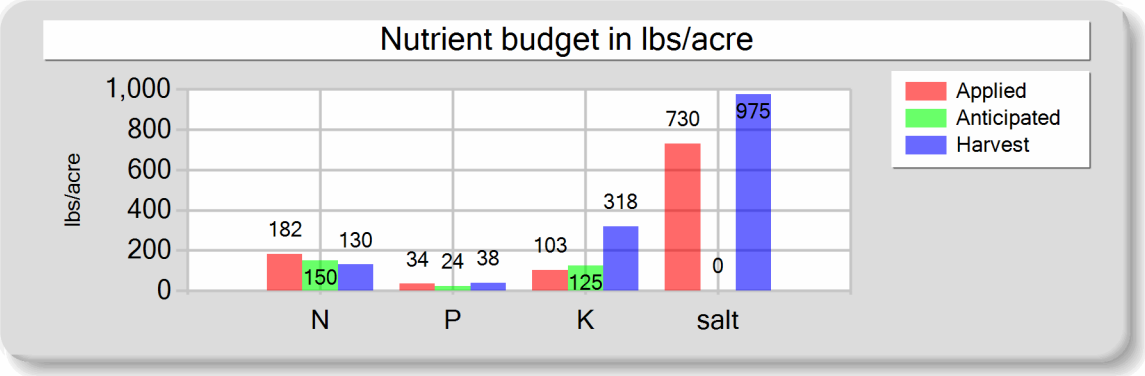
Fresh water applied	
19,706,610.00	*gallons*
725.73	*acre-inches*
29.03	*inches/acre*

Process wastewater applied	
972,000.00	*gallons*
35.80	*acre-inches*
1.43	*inches/acre*

Total harvests for the crop	
1	*harvests*

Sexton - 10/20/2022: Oats, silage-soft dough

Field name: Sexton | Crop: Oats, silage-soft dough | Plant date: 10/20/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	174.92	33.70	102.52	730.46
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	181.92	33.70	102.52	730.46
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	129.60	38.40	318.00	975.07
Nutrient balance	52.32	-4.70	-215.48	-244.61
Applied to removed ratio	1.40	0.88	0.32	0.75

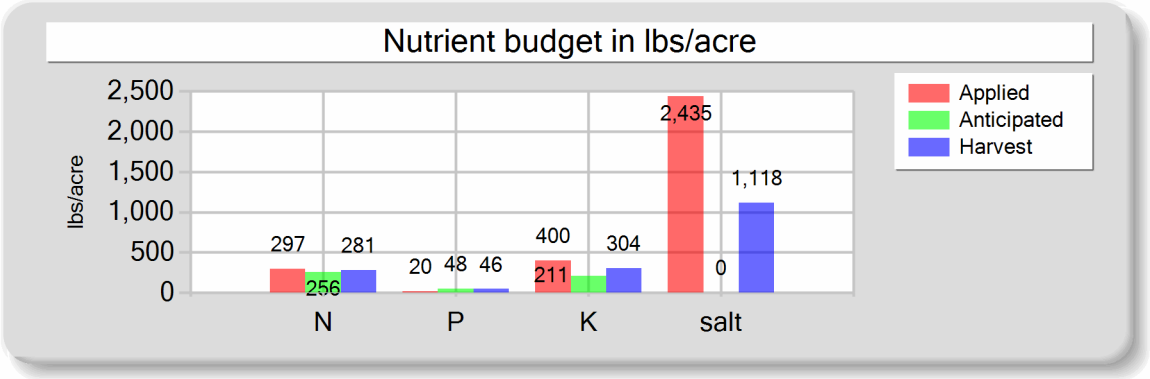
Fresh water applied	
2,975,340.00	*gallons*
109.57	*acre-inches*
4.57	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Sexton - 05/27/2023: Corn, silage

Field name: Sexton | Crop: Corn, silage | Plant date: 05/27/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	8.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	282.49	19.79	400.31	2,050.37
Fresh water	0.00	0.00	0.00	384.93
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	297.49	19.79	400.31	2,435.30
Anticipated crop nutrient removal	256.00	48.00	211.20	0.00
Actual crop nutrient removal	280.82	45.65	303.64	1,118.43
Nutrient balance	16.68	-25.86	96.67	1,316.87
Applied to removed ratio	1.06	0.43	1.32	2.18

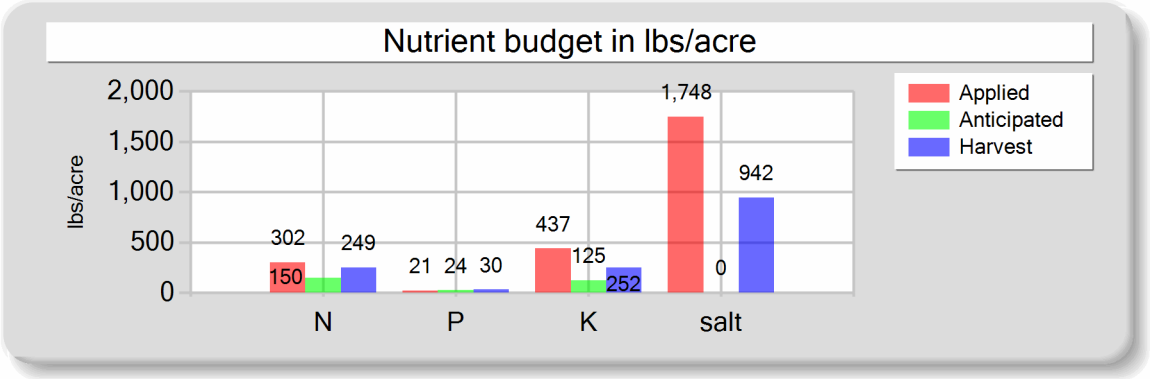
Fresh water applied	
19,086,990.00	*gallons*
702.91	*acre-inches*
29.29	*inches/acre*

Process wastewater applied	
936,000.00	*gallons*
34.47	*acre-inches*
1.44	*inches/acre*

Total harvests for the crop	
1	*harvests*

Shop - 10/23/2022: Oats, silage-soft dough

Field name: Shop | Crop: Oats, silage-soft dough | Plant date: 10/23/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	294.96	20.69	436.97	1,747.90
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	301.96	20.69	436.97	1,747.90
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	249.30	30.00	251.70	941.87
Nutrient balance	52.66	-9.31	185.27	806.03
Applied to removed ratio	1.21	0.69	1.74	1.86

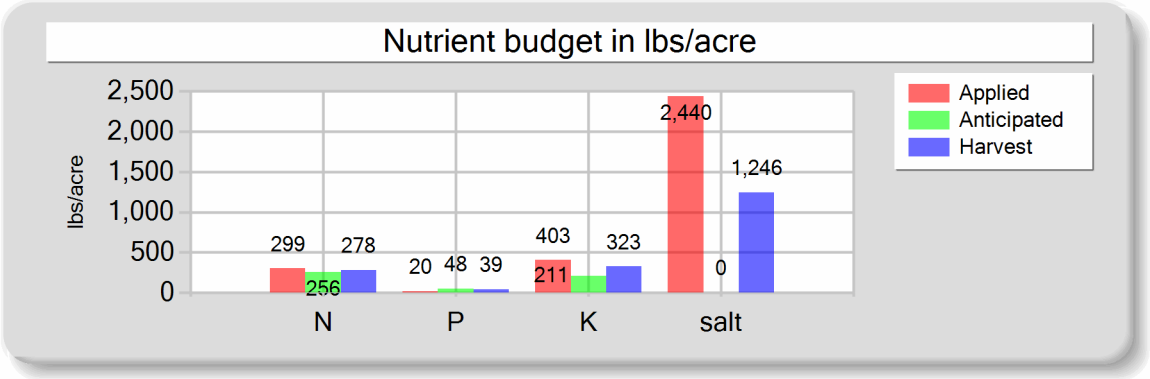
Fresh water applied	
4,200,480.00	*gallons*
154.69	*acre-inches*
4.69	*inches/acre*

Process wastewater applied	
864,000.00	*gallons*
31.82	*acre-inches*
0.96	*inches/acre*

Total harvests for the crop	
1	*harvests*

Shop - 05/25/2023: Corn, silage

Field name: Shop | Crop: Corn, silage | Plant date: 05/25/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	8.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	284.47	19.93	403.11	2,064.70
Fresh water	0.00	0.00	0.00	374.94
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	299.47	19.93	403.11	2,439.64
Anticipated crop nutrient removal	256.00	48.00	211.20	0.00
Actual crop nutrient removal	278.30	39.37	323.10	1,246.25
Nutrient balance	21.17	-19.44	80.01	1,193.39
Applied to removed ratio	1.08	0.51	1.25	1.96

Fresh water applied	
25,563,366.00	*gallons*
941.41	*acre-inches*
28.53	*inches/acre*

Process wastewater applied	
1,296,000.00	*gallons*
47.73	*acre-inches*
1.45	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Corral Solids

Sample and source description: Corral Solids

Sample date: 09/12/2022 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 68.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	30,700.00	10,600.00	27,100.00							17.50
DL	100.00	100.00	100.00							0.01

Separator Solids

Sample and source description: Separator Solids

Sample date: 09/12/2022 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 76.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	27,300.00	5,260.00	16,000.00							11.40
DL	100.00	100.00	100.00							0.01

Corral Solids

Sample and source description: Corral Solids

Sample date: 04/17/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 36.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,400.00	6,700.00	36,400.00							62.00
DL	100.00	100.00	100.00							0.01

Corral Solids

Sample and source description: Corral Solids

Sample date: 04/17/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 36.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,400.00	6,700.00	36,400.00							62.00
DL	100.00	100.00	100.00							0.01

Separator Solids

Sample and source description: Separator Solids

Sample date: 04/17/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 59.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	24,300.00	7,800.00	26,100.00							12.30
DL	100.00	100.00	100.00							0.01

Corral Solids

Sample and source description: Corral Solids

Sample date: 10/09/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 63.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	31,900.00	9,500.00	25,600.00							23.00
DL	100.00	100.00	100.00							0.01

Separator Solids

Sample and source description: Separator Solids

Sample date: 10/09/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 57.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	20,000.00	2,900.00	9,760.00							7.05
DL	100.00	100.00	100.00							0.01

B. PROCESS WASTEWATER ANALYSES

4QWW

Sample and source description: 4QWW

Sample date: 10/24/2022 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1,350.00	943.00	0.00	0.00	94.70	2,000.00								13.70	8,000
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

1QWW

Sample and source description: 1QWW

Sample date: 02/21/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1,010.00	568.00	0.00	0.00	103.00	945.00								8,240.00	4,820
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

2QWW

Sample and source description: 2QWW

Sample date: 04/10/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	913.00	437.00	0.00	0.00	93.20	1,010.00								6,210.00	3,960
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

3QWW

Sample and source description: 3QWW

Sample date: 07/26/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	868.00	557.00	0.00	0.00	60.80	1,230.00								9,440.00	6,300
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

4QWW

Sample and source description: 4QWW

Sample date: 10/30/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1,340.00	815.00	0.00	0.00	150.00	1,660.00								12.20	7,980
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

C. FRESH WATER ANALYSES

Canal

Canal

Fresh Water

Sample description: Fresh Water

Sample date: 08/18/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								89.20	58
DL	0.50	0.50	0.50								1.00	10

D. SOIL ANALYSES

10 Acre

Soils

Sample and source description: Soils

Sample date: 12/05/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			106.00				
DL			4.00				

Heifer

Soils

Sample and source description: Soils

Sample date: 12/05/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			106.00				
DL			4.00				

Hutch

Hutch

Soils

Sample and source description: Soils

Sample date: 12/05/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			112.00				
DL			4.00				

L and M

Soils

Sample and source description: Soils

Sample date: 12/15/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			65.00				
DL			4.00				

Leo's

Soils

Sample and source description: Soils

Sample date: 12/05/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			112.00				
DL			4.00				

Morris

Morris

Soils

Sample and source description: Soils

Sample date: 12/05/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			77.00				
DL			4.00				

Pipe

Soils

Sample and source description: Soils

Sample date: 12/05/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			106.00				
DL			4.00				

Red Barn

Soils

Sample and source description: Soils

Sample date: 12/05/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			117.00				
DL			4.00				

Ricks

Ricks

Soils

Sample and source description: Soils

Sample date: 12/05/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			121.00				
DL			4.00				

Sexton

Soils

Sample and source description: Soils

Sample date: 12/05/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			117.00				
DL			4.00				

Shop

Soils

Sample and source description: Soils

Sample date: 12/05/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			112.00				
DL			4.00				

E. PLANT TISSUE ANALYSES

10 Acre - 10/21/2022: Oats, silage-soft dough

10 Acre - 10/21/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/01/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 72.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	6,780.00	1,270.00	10,800.00		9.44
DL	100.00	100.00	100.00		0.01

10 Acre - 05/29/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/25/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 68.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,080.00	1,140.00	6,070.00		6.60
DL	100.00	100.00	100.00		0.01

Heifer - 10/23/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/01/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 62.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	6,870.00	1,500.00	12,000.00		9.58
DL	100.00	100.00	100.00		0.01

Heifer - 05/25/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/19/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 66.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,470.00	720.00	5,960.00		7.00
DL	100.00	100.00	100.00		0.01

Hutch - 10/22/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/01/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 62.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,410.00	1,490.00	11,800.00		9.89
DL	100.00	100.00	100.00		0.01

Hutch - 05/26/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/19/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,110.00	550.00	5,290.00		6.30
DL	100.00	100.00	100.00		0.01

L and M - 10/23/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/09/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 65.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,460.00	1,190.00	16,200.00		6.98
DL	100.00	100.00	100.00		0.01

L and M - 05/26/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/19/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 66.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,540.00	710.00	6,150.00		6.50
DL	100.00	100.00	100.00		0.01

Leo's - 10/21/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/09/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 65.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,460.00	1,190.00	16,200.00		6.98
DL	100.00	100.00	100.00		0.01

Leo's - 05/31/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/25/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 68.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,050.00	1,020.00	5,960.00		6.70
DL	100.00	100.00	100.00		0.01

Morris - 10/24/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 04/24/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 61.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,700.00	1,440.00	19,400.00		8.32
DL	100.00	100.00	100.00		0.01

Morris - 05/30/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/19/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 66.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,180.00	760.00	4,660.00		6.30
DL	100.00	100.00	100.00		0.01

Pipe - 10/22/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/15/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 67.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	8,610.00	970.00	7,910.00		9.53
DL	100.00	100.00	100.00		0.01

Pipe - 05/27/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/25/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 69.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,440.00	1,130.00	5,380.00		8.00
DL	100.00	100.00	100.00		0.01

Red Barn - 10/22/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/09/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 64.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	6,730.00	1,040.00	14,200.00		12.20
DL	100.00	100.00	100.00		0.01

Red Barn - 05/29/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/19/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 61.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,200.00	910.00	5,350.00		5.40
DL	100.00	100.00	100.00		0.01

Ricks - 10/20/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/15/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 67.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	8,610.00	970.00	7,910.00		9.53
DL	100.00	100.00	100.00		0.01

Ricks - 05/30/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/25/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 68.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,240.00	1,190.00	6,410.00		6.50
DL	100.00	100.00	100.00		0.01

Sexton - 10/20/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/01/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 67.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,320.00	1,280.00	10,600.00		9.97
DL	100.00	100.00	100.00		0.01

Sexton - 05/27/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/19/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,060.00	660.00	4,390.00		5.50
DL	100.00	100.00	100.00		0.01

Shop - 10/23/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 08/15/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 65.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	8,310.00	1,000.00	8,390.00		9.18
DL	100.00	100.00	100.00		0.01

Shop - 05/25/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/19/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 69.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,100.00	580.00	4,760.00		6.00
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

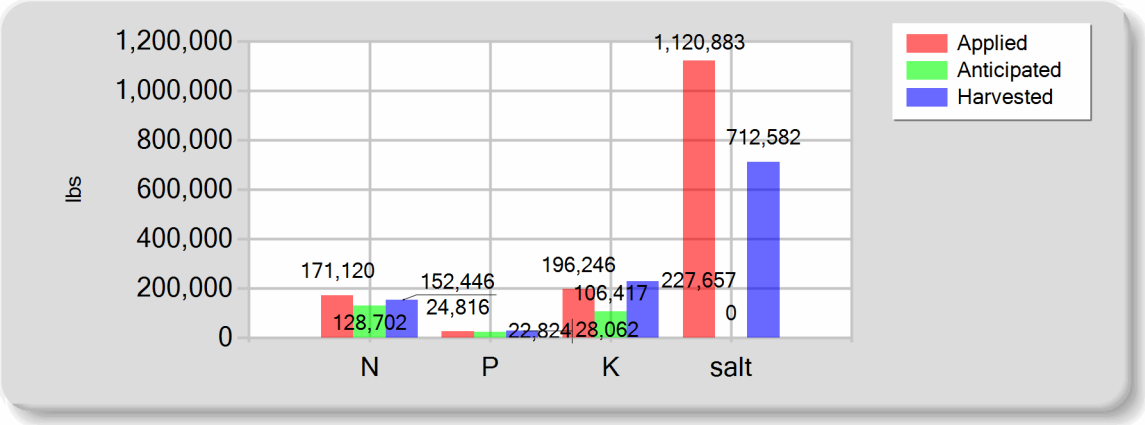
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

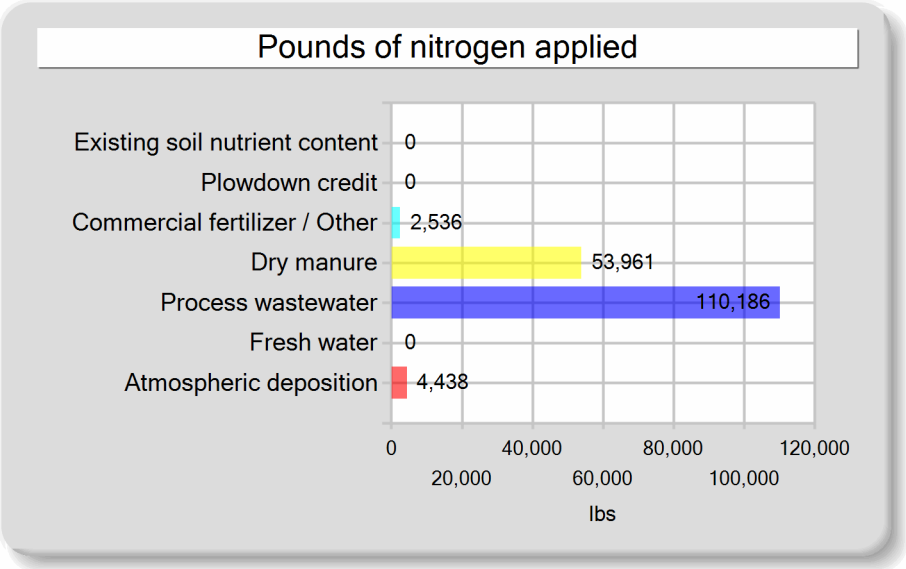
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

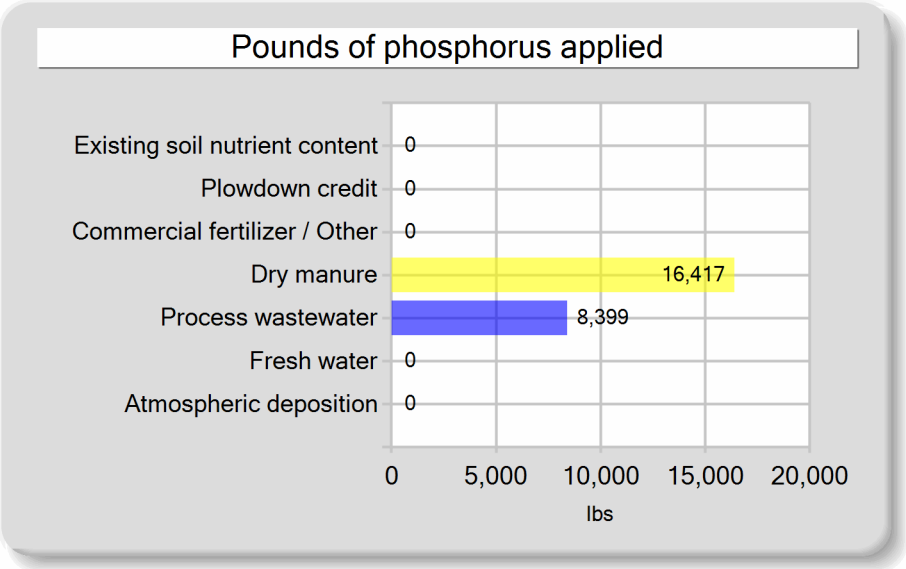
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	2,536.00	0.00	0.00	0.00
Dry manure	53,960.70	16,417.07	48,869.17	276,276.30
Process wastewater	110,185.64	8,398.66	147,377.04	716,681.95
Fresh water	0.00	0.00	0.00	127,924.31
Atmospheric deposition	4,438.00	0.00	0.00	0.00
Total nutrients applied	171,120.34	24,815.73	196,246.21	1,120,882.56
Anticipated crop nutrient removal	128,702.00	22,824.00	106,416.90	0.00
Actual crop nutrient removal	152,446.40	28,062.20	227,656.50	712,582.43
Nutrient balance	18,673.94	-3,246.47	-31,410.29	408,300.13
Applied to removed ratio	1.12	0.88	0.86	1.57

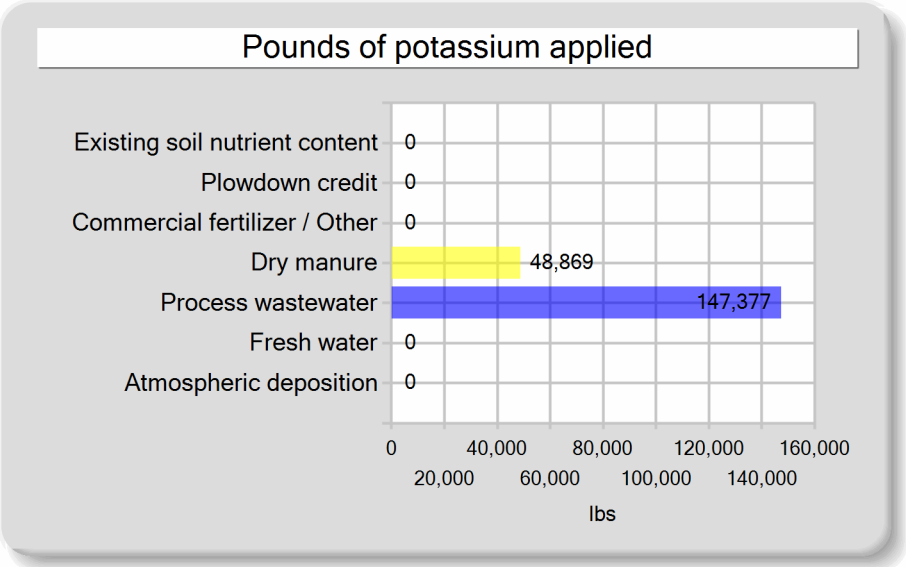
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

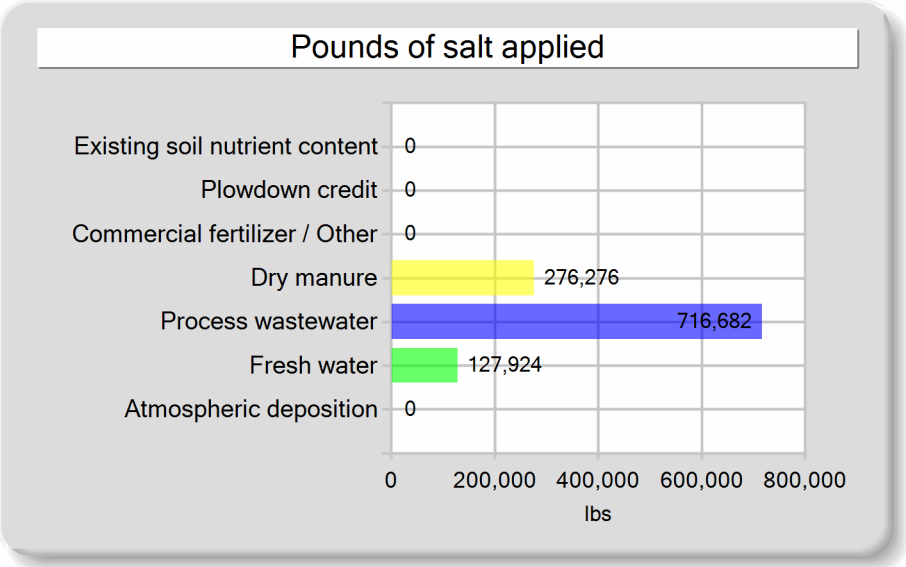


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	Yes
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

No notes entered for this annual report.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Mike Van Vliet (May 1, 2024 18:07 PDT)

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Mike Van Vliet	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
May 1, 2024	
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

ATTACHMENT D

Manure/Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

Instructions:

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, **complete a separate form for each destination.**
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:

Name of Operator: Mike Van Vliet

Name of Dairy Facility: Creekside Dairy

Facility Address: 23234 Lone Tree Rd (Number and Street) Escalon (City) 95320 (Zip Code)

Contact Person Name and Phone Number: Mike Van Vliet (Name) 209 838-1180 (Phone Number)

Manure/Process Wastewater Hauler Information:

Name of Hauling Company/Person: Brasil & Sons

Address of Hauling Company /Person: 21768 E Lone Tree Rd (Number and Street) Escalon (City) 95320 (Zip Code)

Contact Person: Victor Brasil (Name) 209 838-1661 (Phone Number)

Destination Information:

Composting Facility / Broker / Farmer / Other (identify) _____ (please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above): 209

Skiff Ranch (Name) 24323 E Skiff Rd (Number and Street) Escalon (City) 95320 (Zip Code) 607-8343 (Phone Number)

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

Same (Number and Street) _____ (City) _____ (Zip Code) _____ (Assessor's Parcel Number)

Dates Hauled: 10/8/23

Amount Hauled:

Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount:

Manure: 600 Tons / or Cubic Yards (indicate which units used)

Manure Solids Content (if amount reported in tons): 57.2 % Moisture

Manure Density (if amount reported in cubic yards): _____

Method used to determine amount of manure: _____

60 Loads times 10 Ton per Load

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: _____ Gallons

Method used to determine volume of process wastewater: _____

_____

Written Agreement:
Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use? (please check one)

____ Yes _____ No

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after **31 December 2007** to such party.
_____ (Operator shall provide initials here to acknowledge this requirement).

Certification:
I declare under the penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator's Signature: Mike Volet Date: 4-20-24

Hauler's Signature: Victor Brasil Date: 10-10-23

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
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40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

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Compliance Analysis Report

Customer
Creekside Dairy
23234 Lone Tree Road
Escalon, CA 95320

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 12/15/23 11:30 AM
SUBMITTED BY: Jeannette
ANALYZED DATE: 12/19/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature above qualifying at 10.4 C

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Barn	T3349004A	Electrical Conductivity (EC)	EPA 120.1 and	0.546	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	11.9	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	350	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	94.4	98.8 (T3349002A)	100 (T3349002A)	1.20
Electrical Conductivity (EC)	99.0			0.033
Nitrate Nitrogen (NO3-N)	105	107 (T3346026B)	105 (T3346026B)	1.40

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

ELAP Certificate No.: 2714

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Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 12/15/23 11:30 AM
SUBMITTED BY: Jeannette
ANALYZED DATE: 12/19/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature above qualifying at 11.2 C

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
CLL	T3349004B	Electrical Conductivity (EC)	EPA 120.1 and	0.910	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	26.3	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	582	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	94.4	98.8 (T3349002A)	100 (T3349002A)	1.20
Electrical Conductivity (EC)	99.0			0.033
Nitrate Nitrogen (NO3-N)	105	107 (T3346026B)	105 (T3346026B)	1.40

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Grower

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F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 12/15/23 11:30 AM
SUBMITTED BY: Jeannette
ANALYZED DATE: 12/19/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature above qualifying at 11.0 C

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Leo's	T3349004C	Electrical Conductivity (EC)	EPA 120.1 and	0.885	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	18.6	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	566	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	94.4	98.8 (T3349002A)	100 (T3349002A)	1.20
Electrical Conductivity (EC)	99.0			0.033
Nitrate Nitrogen (NO3-N)	105	107 (T3346026B)	105 (T3346026B)	1.40

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Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 12/15/23 11:30 AM
SUBMITTED BY: Jeannette
ANALYZED DATE: 12/19/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature above qualifying at 10.6 C

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Mike's	T3349004D	Electrical Conductivity (EC)	EPA 120.1 and	1.12	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	40.4	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	718	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	94.4	98.8 (T3349002A)	100 (T3349002A)	1.20
Electrical Conductivity (EC)	99.0			0.033
Nitrate Nitrogen (NO3-N)	105	107 (T3346026B)	105 (T3346026B)	1.40

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Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 12/15/23 11:30 AM
SUBMITTED BY: Jeannette
ANALYZED DATE: 12/19/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature above qualifying at 10.8 C

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Pit	T3349004E	Electrical Conductivity (EC)	EPA 120.1 and	1.01	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	24.3	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	648	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	94.4	98.8 (T3349002A)	100 (T3349002A)	1.20
Electrical Conductivity (EC)	99.0			0.033
Nitrate Nitrogen (NO3-N)	105	107 (T3346026B)	105 (T3346026B)	1.40

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Escalon, CA 95320

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 12/15/23 11:30 AM
SUBMITTED BY: Jeannette
ANALYZED DATE: 12/19/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature above qualifying at 10.2 C

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
RV	T3349004F	Electrical Conductivity (EC)	EPA 120.1 and	0.887	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	26.4	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	568	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	94.4	98.8 (T3349002A)	100 (T3349002A)	1.20
Electrical Conductivity (EC)	99.0			0.033
Nitrate Nitrogen (NO3-N)	107	113 (T3349029A)	111 (T3349029A)	1.55

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Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 12/15/23 11:30 AM
SUBMITTED BY: Jeannette
ANALYZED DATE: 12/19/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature above qualifying at 10.4

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Shop	T3349004G	Electrical Conductivity (EC)	EPA 120.1 and	1.02	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	33.8	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	655	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	94.4	98.8 (T3349002A)	100 (T3349002A)	1.20
Electrical Conductivity (EC)	99.0			0.033
Nitrate Nitrogen (NO3-N)	107	113 (T3349029A)	111 (T3349029A)	1.55

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ELAP Certificate No.: 2714

DENELE ANALYTICAL, INC.

865 S. KILROY ROAD
TURLOCK, CA 95380
PH# 209-634-9055 · FAX# 209-634-9057

DAIRY COMPLIANCE

Client: Creekside
Mailing address:
(not facility address)
Phone#:

Authorized Copy Released To: F&R AG SERVICES
Sampled By: Trevor
Project ID:

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
H1 EC*, NO3-N, NH4-N*
H6 EC, TN, TDS
H7 Combination of H1 & H6
H14 EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
H3 EC*, NH4-N, TKN, TP, K, TDS
H4 EC*, NO3-N, NH4-N, TKN, TP, K, TDS
H5 Ca, Mg, Na, Bicarb, Carb, SO4, Cl
H8 Combination of H3 &H5
H9 Combination of H4 &H5

TILE DRAIN
H2 EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*
____ (Check here if performed)

DISCHARGE WATER
E1 EC*, Temp*, pH*, NO3-N, NH3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
E2 EC,* Temp*, pH*, NO3-N, NH3-N*, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
E3 EC*, Temp*, pH*, NO3-N, NH3-N*, NH4-N*, TP, Total & Fecal Coliforms
E4 EC*, Temp*, pH*, NO3-N, NH3-N*, NH4-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
M1 % Moisture
M3 TN TP, K, % Moisture
M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
M7 Combination of M3 & M4

PLANT TISSUE
F18 % Moisture, Ash, TN, P, K, DM and as received
F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
F19+ % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
K3 Soluble Phosphorus
K4 1ft NO3, Organic Matter
K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
K7 Combination of K3 & K4
K8 Combination of K3 & K5

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	H1	H3	M3	F18	OTHER
1	12/15/23		Barn	Domestic	10.4		✓			✓	
2			CU		11.2						
3			Leo's		11.0						
4			Mike's		10.6						
5			Pit		10.8						
6			RV		10.2						
7			Shop		10.4						
8											
9											
10											

Analysis

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time	Logged in By:	LABORATORY NUMBER
	Jeanette Sawyer	F&R AG SERVICES				
2nd		DENELE ANALYTICAL, INC. (NO)	12-15-23 @ 11:30			T3349004

(LABORATORY USE ONLY)

2023 Annual Report Creekside

Final Audit Report 2024-05-02

Created:	2024-05-01
By:	Vince Furtado (vfurtado@fragservices.com)
Status:	Signed
Transaction ID:	CBJCHBCAABAA4m_hSRnm7wzXGSxelUC2otAwBBB35tXG

"2023 Annual Report Creekside" History

Document created by Vince Furtado (vfurtado@fragservices.com)
2024-05-01 - 11:22:25 PM GMT

Document emailed to trevorvanvliet@yahoo.com for signature
2024-05-01 - 11:23:36 PM GMT

Email viewed by trevorvanvliet@yahoo.com
2024-05-02 - 1:04:31 AM GMT

Signer trevorvanvliet@yahoo.com entered name at signing as Mike Van Vliet
2024-05-02 - 1:07:30 AM GMT

Document e-signed by Mike Van Vliet (trevorvanvliet@yahoo.com)
Signature Date: 2024-05-02 - 1:07:32 AM GMT - Time Source: server

Agreement completed.
2024-05-02 - 1:07:32 AM GMT

Adobe Acrobat Sign